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Task Summary

A simple rule-based conversational chatbot built in Python. The chatbot asks the user's name, then runs an interactive loop that reads and cleans user input, recognizes greetings and several topics using if/elif/else and the **in** operator for flexible matching, and exits cleanly when the user types 'bye' or 'exit'.

Requirements Checklist

#	Requirement	Met?
1	Ask the user's name before the main conversation	Yes
2	Use while True: to keep the conversation running	Yes
3	Use input() to receive messages	Yes
4	Clean the input using .lower().strip()	Yes
5	Recognize at least two greetings (hello, hi)	Yes
6	Answer at least three different questions/topics	Yes (5 topics)
7	Use if / elif / else for decision making	Yes
8	Use break to exit on bye / exit	Yes
9	Use the user's name in at least one response	Yes
10	Useful response for unknown messages (else)	Yes
Challenge	Use 'in' operator for flexible greeting matches	Yes

Python Source Code (chatbot.py)

```
"""
Rule-Based Python Chatbot
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A simple rule-based conversational chatbot that:
 - asks the user's name before the main conversation
 - keeps chatting using while True:
 - reads input with input()
 - cleans input using .lower().strip()
 - recognizes greetings, questions/topics, and an exit command
 - uses if / elif / else for decision making
 - uses the in operator so variations like "hello bot" still work
"""

def main():
    # 1. Ask the user's name before the main conversation
    name = input("Chatbot: Hi there! What's your name? ").strip()
    print(f"Chatbot: Nice to meet you, {name}! Type 'bye' or 'exit' whenever you want to stop.\n")

    # 2. Keep the conversation running
    while True:
        user_input = input(f"{name}: ")

        # 3 & 4. Clean the input
        message = user_input.lower().strip()

        # 8. Exit condition
        if message in ("bye", "exit"):
            print(f"Chatbot: Goodbye, {name}! It was nice talking to you.")
            break

        # 5. Greetings (at least two, flexible with 'in')
        elif "hello" in message or "hi" in message:
            print(f"Chatbot: Hello {name}! How can I help you today?")

        # 6. Topic 1 - how are you
        elif "how are you" in message:
            print("Chatbot: I'm just a program, but I'm running great, thanks for asking!")

        # 6. Topic 2 - what is your name
        elif "your name" in message:
            print("Chatbot: I'm a simple rule-based chatbot built in Python.")

        # 6. Topic 3 - what can you do / help
        elif "what can you do" in message or "help" in message:
            print(f"Chatbot: I can chat with you, {name}! Try asking about the time, "
                  f"my name, or how I'm doing.")

        # Extra topic - time
        elif "time" in message:
            print("Chatbot: I can't check a clock, but it's always a good time to code!")

        # Extra topic - thanks
        elif "thank" in message:
            print(f"Chatbot: You're welcome, {name}!")

        # 10. Fallback for unknown messages
```

```
        else:
            print("Chatbot: Sorry, I didn't quite understand that. Could you rephrase it?")


if __name__ == "__main__":
    main()
```

Test Run Transcript

Below is an actual run of the program with the required test inputs: hello, HELLO, hello, hi, a question (how are you), an unknown message, and finally bye.

```
Chatbot: Hi there! What's your name? Rama
Chatbot: Nice to meet you, Rama! Type 'bye' or 'exit' whenever you want to stop.

Rama: hello
Chatbot: Hello Rama! How can I help you today?
Rama: HELLO
Chatbot: Hello Rama! How can I help you today?
Rama: hello
Chatbot: Hello Rama! How can I help you today?
Rama: hi
Chatbot: Hello Rama! How can I help you today?
Rama: how are you
Chatbot: I'm just a program, but I'm running great, thanks for asking!
Rama: qwertyxyz
Chatbot: Sorry, I didn't quite understand that. Could you rephrase it?
Rama: bye
Chatbot: Goodbye, Rama! It was nice talking to you.
```

Observation: 'hello' and 'HELLO' both trigger the greeting because the input is lowercased before matching. The unknown message falls through to the else branch, and typing 'bye' triggers the break statement, ending the loop.